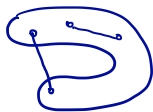
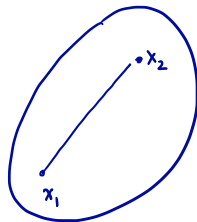


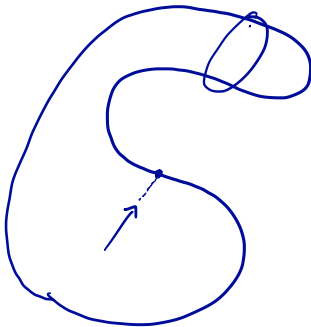
Convex Sets - definition

A set $C \subseteq \mathbb{R}^n$, is convex
if $\forall x_1, x_2 \in C, \forall \lambda \in [0, 1]$

$$\lambda x_1 + (1 - \lambda)x_2 \in C$$



Why important



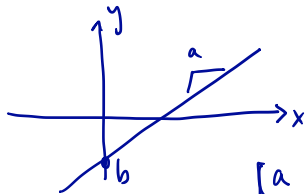
Subspace and affine space

span (range) vs null space definitions

Vectors: $v_1, \dots, v_k$ $V = \text{span} \{v_1, \dots, v_k\} = \left\{ v = \sum_{i=1}^k \alpha_i v_i, \alpha_i \in \mathbb{R} \right\}$

$$V = \text{Range}(A), \quad A = \begin{bmatrix} | & & | \\ v_1 & \dots & v_k \\ | & & | \end{bmatrix}$$

⑧ $\text{NullSpace}(M) = \left\{ x : Mx = 0 \right\}$



$$y = ax + b$$

$$-b = a \cdot x + -1 \cdot y$$

$$\begin{bmatrix} a & -1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -b$$

Exercise: Check for yourselves that
Range & NullSpace
are closed under scaling
and addition.

Half space / hyperplane - definition and convexity

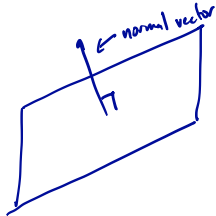
normal vector and offset

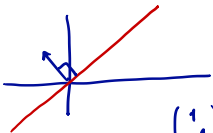
In 2-d ($\mathbb{R}^2$), a line is defined by: $\underline{s^T x = b}$
(In prev slide: $s = \begin{pmatrix} a \\ -1 \end{pmatrix}$, $x = \begin{pmatrix} x \\ y \end{pmatrix}$, $b = -b$)

In n-d ($\mathbb{R}^n$) this defines: a

$\downarrow$
 $\{x \in \mathbb{R}^n : s^T x = b\}$ - Hyperplane - "co-dimension 1" space

$\uparrow$
normal vector



Ex:  $y = 1 \cdot x + 0$
 $\begin{pmatrix} 1 \\ -1 \end{pmatrix}^T \begin{pmatrix} x \\ y \end{pmatrix} = 0$

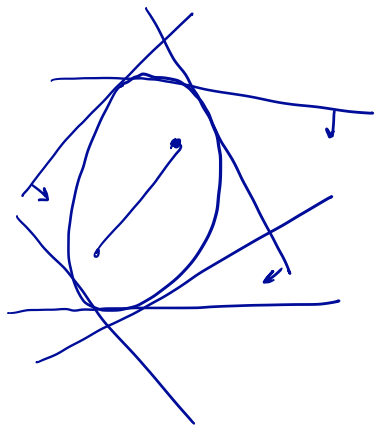
The diagram shows a 2D Cartesian coordinate system with a red line passing through the origin. A vector labeled "normal vector" with an arrow points from the origin perpendicular to the red line. The equation $y = 1 \cdot x + 0$ is written to the right of the graph. Below the graph, the vector equation $\begin{pmatrix} 1 \\ -1 \end{pmatrix}^T \begin{pmatrix} x \\ y \end{pmatrix} = 0$ is written.

Half spaces are fundamental: duality

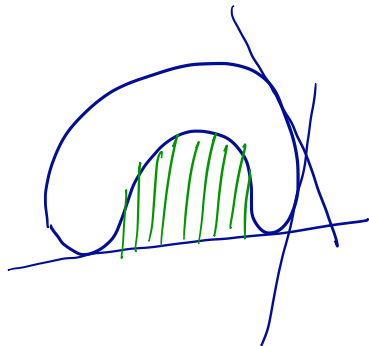
$$\text{Hyperplane: } H_{s,b} = \{x \in \mathbb{R}^n : s^T x = b\}$$

$$H_{s,b}^+ = \{x \in \mathbb{R}^n : s^T x \geq b\},$$

$$H_{s,b}^- = \{x \in \mathbb{R}^n : s^T x \leq b\}$$



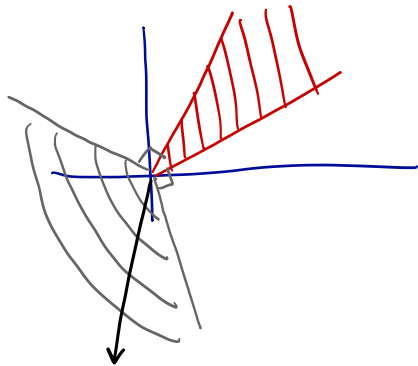
vs



Cones and polar cones (1/2)

K is called a cone if:

$$x_1, x_2 \in K \Rightarrow \alpha_1 x_1 + \alpha_2 x_2 \in K \quad \forall \alpha_1, \alpha_2 \geq 0$$



Polar cone:

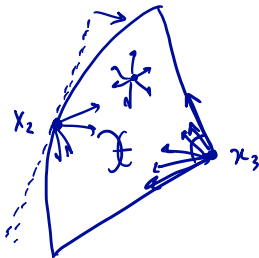
$$K^0 = \left\{ z : \underbrace{\langle z, x \rangle}_{\leq 0}, \forall x \in K \right\}$$

Cones and polar cones (2/2)

We will use these to
characterize optimality.

Tangent cone

$$\begin{aligned} \min: & f(x) \\ \text{st: } & x \in \mathcal{X} \end{aligned}$$



$T_{\mathcal{X}}(x)$: If x is an internal point of $\mathcal{X}$
i.e., $x \in \text{int } \mathcal{X}$, then $T_{\mathcal{X}}(x) = \mathbb{R}^n$

For $x = x_2$: $T_{\mathcal{X}}(x_2)$ = indicated half-space.

For $x = x_3$: $T_{\mathcal{X}}(x_3)$ = cone defined by two bdy vectors.

Normal cone

Polar of Tangent cone :

$$N_{\mathcal{X}}(x_1) = \{0\}$$

$$N_{\mathcal{X}}(x_2) =$$

